Assignment 3

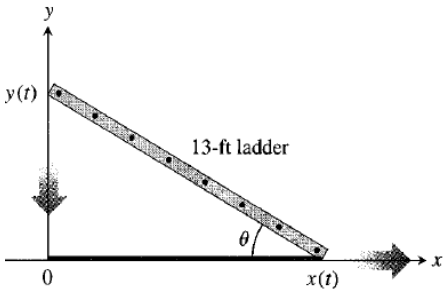
MA-110 Calculus and Analytical Geometry

Fall – 2024

Apply the concepts of differentiation to solve the following problems.

Q 1. **A sliding ladder.**

A 13-ft ladder is leaning against a house when its base starts to slide away. By the time the base is 12 ft from the house, the base is moving at the rate of 5 ft/sec.



a) How fast is the top of the ladder sliding down the wall then?
b) At what rate is the area of the triangle formed by the ladder, wall, and ground changing then?
c) At what rate is the angle θ between the ladder and the ground changing then?

Q 2. **Extremes of Given Function**

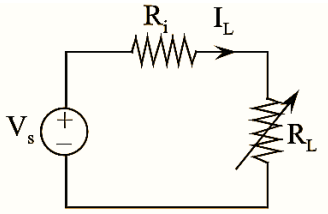
a) Find the intervals in which the function $g(x) = x^2\sqrt{5-x}$ is increasing and decreasing.

b) Then identify the function's local extreme values, if any.

c) Using the information derived in (a) and (b), sketch the graph of the function.

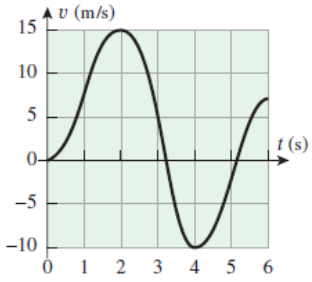
Q 3. Maximum Power Transfer

A voltage source is supplying current to a load resistor R_L, through a resistor $R_i = 100\Omega$, as shown in figure. Here R_L is adjustable and it is required to adjust R_L such that power delivered to the load is maximum. Find the value of R_L, satisfying this requirement.



Q 4. Rectilinear Motion

The accompanying figure shows the graph of velocity versus time for a particle moving along a straight line.



(a) Identify the time instants when the particle is not moving.
(b) Identify the time instants when the particle is not accelerating.
(c) Identify the time intervals when the particle is (i) accelerating (ii) decelerating.
(d) Identify the time instants when the particle changes its direction.
(e) Based on the results found above, draw rough sketches of the graphs of (i) speed versus time (ii) acceleration versus time

Note: Your submitted assignment must be your own effort. Copying from a friend, solution manual, internet or any other source may lead you to F grade. Sharing your solution with others is equally punishable.

Due Date: 13th Dec. 2024